

The Immersive Art Therapy Driven by AIGC: An Innovative Approach to Alleviating Children's Nyctophobia

Jinlin Miao*
School of Industrial Design
China Academy of Art
Hangzhou, China
Design Anything Lab
China Academy of Art
Hangzhou, China
miaojinlinchloe@gmail.com

Zhiyuan Zhou*
School of Design & Innovation
China Academy of Art
Hangzhou, China
Design Anything Lab
China Academy of Art
Hangzhou, China
3210601057@caa.edu.cn

Yilei Wu
School of Industrial Design
China Academy of Art
Hangzhou, China
Design Anything Lab
China Academy of Art
Hangzhou, China
1223145658@qq.com

Fenggui Rao†
The Hong Kong Polytechnic
University
Hong Kong, China
Design Anything Lab
China Academy of Art
Hangzhou, China
admin@whitesir.cn

Fanjing Meng†
University College London
London, United Kingdom
Design Anything Lab
China Academy of Art
Hangzhou, China
mengfanjing2024@163.com



Figure 1

Abstract

This study presents a participatory co-creation installation based on Artificial Intelligence Generated Content (AIGC) for alleviating children's Nyctophobia through art therapy. By conducting interviews with parents, children, and psychologists, the study analyzes the causes of nyctophobia and proposes a solution that integrates intelligent technologies. The study employs real-time interactive technologies to transform children's behaviors into positive visual experiences, aiding them in building a sense of security in dark environments. The study's contribution lies in proposing a design framework that integrates serious games and exposure therapy,

offering new perspectives for psychological interventions targeting children's mental health.

CCS Concepts

• **Human-centered computing** → *Hypertext / hypermedia*.

Keywords

Art therapy, dark phobia, AI-generated content

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1 Introduction

Phobias are a common among children and adolescents [12, 19, 24], with Nyctophobia affecting 10 percent to 15 percent of reported treatment cases [20]. Severe nighttime fears and sleep disturbances

*These authors contributed equally to this work.

†Co-corresponding authors.

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significantly impact 20 percent to 30 percent of children [3], disrupting their daily lives and potentially leading to anxiety disorders, thereby amplifying the scope and severity of their impact [22]. Recent studies suggest that children have a positive attitude toward games, making serious games a promising and innovative platform for psychological interventions [9]. Compared to traditional methods, serious games demonstrate unique advantages in psychological interventions due to high immersion and flexibility [10]. Cognitive behavioral therapy (CBT) techniques, such as exposure therapy, have proven effective in alleviating phobias [25], but often lack personalization and interactivity. The rapid emergence of generative artificial intelligence (AIGC) opens new possibilities for creating adaptive and immersive intervention tools. AIGC can generate dynamic, real-time feedback based on children's behavior, turning actions into positive interactive experiences. This study addresses the question: How can serious games and exposure therapy be combined with AIGC to design an interactive game for alleviating children's nyctophobia? Using a combined qualitative and design research approach, in-depth interviews with children experiencing nyctophobia and their parents explored its causes and symptoms, forming the basis for design requirements. Guided by exposure therapy principles, an immersive interactive game using AIGC was developed. The game provides real-time motion capture and transforms children's movements into positive visual feedback, fostering safety and positive associations in dark environments. This study proposes a design framework for integrating AIGC into serious games, offering an innovative intervention tool to alleviate children's fear of the dark. It explores the theoretical foundation, design methodology, and effectiveness of this approach through literature review, interviews, design development, and evaluation, contributing valuable insights into AIGC-based mental health interventions for children.

2 Related work/Background Research

Applications of Gamified Interaction in Psychological Therapy: Interactive games have become an effective psychological therapy tool for children [5]. For example, Starrypia [6] uses AR, music therapy, and sensory integration to help children with autism improve focus, while EMooly [18] utilizes AR and GenAI to enhance social-emotional learning for autistic children. These gamified interventions have been shown to boost engagement and effectiveness compared to traditional methods. However, the impact of immersion and engagement in therapeutic environments, key factors influencing outcomes, remains underexplored.

Existing Methods for Treating Childhood Nyctophobia: Many studies have explored treatments for nyctophobia [1, 3]. For example, Elizabeth Cornwall et al. [7] used emotional imagery therapy to treat nyctophobia in children aged 7 to 10, resulting in significant improvements in symptoms. NoPhobiar [32] emphasized the importance of childlike scenarios and virtual fear representations in therapeutic games. However, these studies did not incorporate exposure therapy, which is proven to be one of the most effective and safe treatments for specific phobias [17].

Interactive Drawing with GenAI for Therapeutic Purposes: Interactive drawing has been explored for physical rehabilitation and emotional therapy due to its therapeutic and expressive benefits [2]. When combined with GenAI, these activities can improve

Serial number	Gender	Age	Experience of nyctophobia	Identity
P1	M	21	YES	brother
P2	F	21	YES	sister
P3	M	32	NO	Parents
P4	M	22	YES	brother
P5	F	54		Psychological consultant
P6	M	7	YES	Children
P7	F	7	YES	Children
P8	F	24		Psychological consultant
P9	F	51		Psychological consultant
P10	M	36		Art therapy consultant

Table 1: Interview Participant Demographics

therapy outcomes [11]. However, the integration of GenAI with interactive drawing is still rare. For instance, Baron et al. developed a VR drawing game for upper-limb therapy without GenAI [2], and Boadi et al. explored guided interactive art for emotional regulation without GenAI [4].

3 User study

3.1 Interview Methods

We conducted semi-structured interviews with ten participants, including patients, family members, three psychologists and an art therapy expert, recruited via a contact list from previous studies and social media. Participants had personal or parental experience with nyctophobia, and the psychologists along with the art therapy expert had extensive experience treating children's psychological issues. All participants were informed about the study's purpose and gave their consent. Table 1 summarizes participants' backgrounds, with ages ranging from 7 to 51. The 20-30 minute online interviews covered experiences with nyctophobia, its causes and symptoms, the role of interactive technologies, and the potential of art therapy at home. Follow-up questions explored notable points, providing valuable insights for design decisions.

3.2 Thematic analysis

We used thematic analysis to process the data. Interviews in Mandarin were recorded, transcribed, or documented during sessions. Collaborators independently reviewed and coded the transcripts, with cloud-based tools aiding data storage and analysis. Open coding generated initial codes, followed by discussions to finalize themes. Key themes included the therapeutic value of creative activities, fear of enclosed spaces and loneliness, and the sense of safety from glowing objects. Three core themes emerged to guide design principles: Integration of creativity and emotional regulation; Enhancing immersion to increase children's engagement; The potential of art therapy as a creative and therapeutic activity.

3.2.1 The Integration of Creativity and Emotional Regulation. Light sources evoke a sense of safety, as one participant noted: "It's best to

have an element of fire." (P1). "When I was scared, I took a flashlight and buried my head in the quilt." (P7). However, ghost stories and media often amplify fear, as seen in one participant's description: "The ceiling, walls, and floor are all dark red, like a haunted hotel" (P2). "It seems like there are ghosts. I feel something is going on after turning off the lights." (P6). "I think of many, many ghosts." (P7). We propose that design principles should focus on integrating creativity and emotional regulation, harnessing children's imaginative fears of darkness and the creativity within the drawing process.

3.2.2 Enhancing Immersion and Child Engagement. P1 shared that she was once afraid of dark, enclosed spaces: "The room is very dark, great for sleeping, but when I was younger, I was scared" (P1). "A light-opening door would be better." (P7). Uncertainty also causes fear: "With the light on, everything's visible and less frightening, but when it's off, there's uncertainty" (P2). Loneliness adds to the fear: "If I have someone with me, I'm not afraid, but I get scared when alone" (P2). These insights suggest that the design should foster a strong sense of space while offering companionship to encourage children's active participation in art therapy.

3.2.3 Art Therapy as a Creative Healing Activity with Great Potential. Interviews revealed that while parents value the benefits of creative activities, they are less familiar with art therapy. P3 noted: "Family-based art therapy feels too demanding; we're tired from work, and grandparents find it challenging. It's easier to send the child to paint at an institution." Accessibility is also a concern: "I haven't heard of art therapy" (P3). This suggests that despite its benefits, art therapy remains limited in reach due to perceived challenges and barriers.

4 Design Implementation

This design has been exhibited in many cities (Beijing, Hangzhou, Shenzhen, Guangshan, etc.), and tens of thousands of participants have visited or interacted with the installation. During this period, the project reflects the user friendliness and system stability.

4.1 Design Rationale

4.1.1 Integration of Creativity and Emotional Regulation. Research has shown that creative activities, especially therapeutic art creation, can effectively support emotional regulation [8]. This study integrates interactive drawing with AIGC, allowing physical movements to generate virtual drawing content rapidly. This approach enhances children's sense of accomplishment and creative enjoyment [14], offering a dynamic environment for self-expression to relieve stress and regulate emotions.

4.1.2 Enhancing Immersion and Children's Engagement. Interactive games have proven therapeutic benefits for children by providing engaging and immersive experiences [15, 21]. Compared to VR, projection technology is cost-effective, accessible, and integrates seamlessly into physical spaces, offering a freer experience without device constraints [16]. This design adopts a full-space interactive concept, using three-wall projections to create strong immersion.

4.1.3 Exposure Therapy and Stylized Monster Design. Exposure therapy effectively treats childhood phobias by helping children gradually confront fear-inducing objects in a controlled setting

[17]. This study applies exposure therapy to address nyctophobia by simulating dark environments. Flowing particle effects create a calming atmosphere to reduce anxiety, while common fear triggers, such as monsters identified in interviews, are integrated as cartoon-style drawing elements to minimize their fear-inducing nature. This design enhances children's sense of safety and engagement during therapy [26].

4.2 Prototype Game Design and Features

4.2.1 Data collection and Diffusion models training. To generate child-like artwork, we invited over 400 children aged 7 to 12 to create drawings, collecting 423 samples as training data for the Diffusion model. After processing the images with Brime for cropping, we extracted the main elements and style prompts using the DeepBooru plugin for Stable Diffusion and stored them in txt files. Next, we trained the LoRA model using Kohya's Stable Diffusion trainers [13], creating a model that generates images in a style suitable for children's drawings, which we then used for image generation during user interaction.

4.2.2 System Construction. Our system consists of a local Python program, a TouchDesigner module, a cloud-based Stable Diffusion server, and spatial mapping with multi-wall projections. **Cloud Component:** We utilized AutoDL to deploy Stable Diffusion on rented GPU (RTX 4090D) and CPU (15-core Intel Xeon Platinum 8474C) instances. A pre-trained children's art-style LoRA model was set as the default generation model. **Local Component:** TouchDesigner created flowing particle effects using SOP nodes, while JSON enabled communication with the cloud to receive generated images and user data. A Python program integrated all system components. **Workflow:** Users interact with ArUco markers, each linked to prompts that guide Stable Diffusion to generate images. A Python program captures the scene, detects the markers, and extracts their coordinates, which are converted into trajectory curves and JSON data for cloud processing. The cloud server generates images and returns them to the local system, where OpenCV GrabCut removes the background. The processed images are then displayed via TouchDesigner. Marker coordinates are also sent in real-time via UDP to TouchDesigner, enabling particle generation and movement, enhancing the interactive experience [23]. **Spatial Mapping:** To achieve an immersive experience beyond traditional screens, we used the open-source software Spout to stream real-time content from TouchDesigner to Resolume Arena for slicing and transformation. The processed visuals are then mapped onto two vertical walls and the floor, creating a surround-view perspective and an immersive environment.

4.3 Design Evaluation

We interviewed three psychology experts (P5, P8, P9) and an art therapy expert (P10), presenting the art installation through design diagrams and a demonstration video. They all affirmed our current design: "I think your overall direction is quite good, and it has covered many important parts within the framework of art therapy." (P8) "It may have a future. I think it can be very helpful." (P9) "So you will find that art is a good way to simulate and then replay. It has such an advantage that it is more visually intuitive and may be more interesting, and it is not a very social prospect experiment."

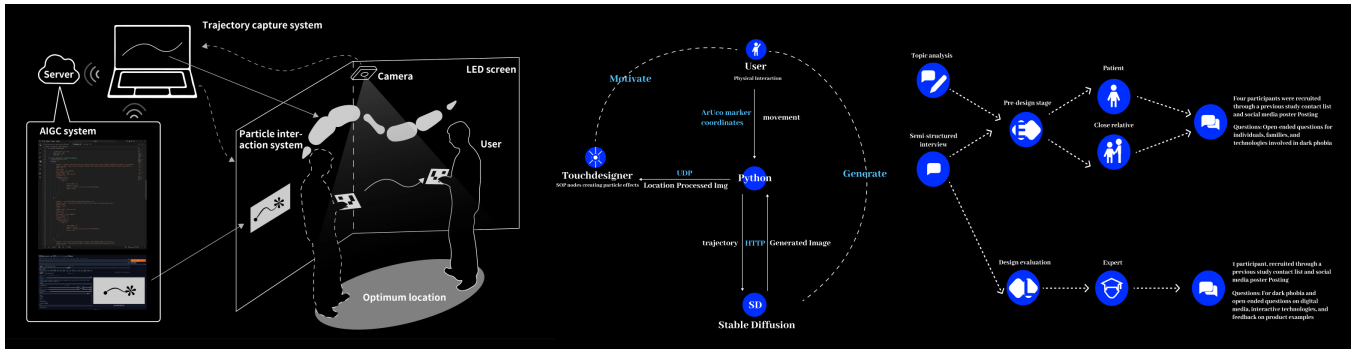


Figure 2: An image of the system

(P10) The experts recommended a gradual increase in darkness with supportive interventions tailored to individual needs: "Exposure therapy must be safe and progressive." (P5) "I think using CBT is a bit too much." (P8) They also emphasized the importance of parental involvement, as fear causes are often personal. The experts gave some suggestions for follow-up experimental evaluation: "I assume that if he and his parents go to such an interactive situation, suppose you have three interactions, right? Then you ask the parents to go in with them first, three people—the three parents go in together—and then if you give them a choice in the three interactive situations, for example, halfway through, you ask the child to make a decision: in the remaining environment, do you want to stay alone, or do both parents accompany you, or do one of the parents accompany you? If you find that the children make different choices in these three modes, especially in one of the interactive modes, where the children are willing to stay alone, then it means that the effect is better." (P10) Additionally, the expert highlighted AI's potential to expand access to therapy and create intuitive visual experiences for children struggling with meditation-based interventions.

5 CONCLUSION

Existing research predominantly focuses on traditional psychological treatment methods, such as cognitive behavioral therapy and exposure therapy, while overlooking the potential of emerging technologies to enhance the interactivity and personalization of interventions. By examining children's nyctophobia, this study addresses the limitations of current approaches in terms of personalization, interactivity, and immersion. A solution is proposed that integrates serious games, exposure therapy, and generative AI (AIGC) to lower the barriers to intervention and enhance children's engagement and sense of security through real-time interactive technologies. This framework not only provides theoretical support and practical references for future studies but also opens new avenues for advancing mental health research in the HCI field.

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